

GreenDev AI - Sustainability Report

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1. Code Metrics

Functions	1
Loops	2
Nested Loops	1
Lines of Code	8
Complexity	4
Task Type	matrix-multiply
Recursion	False

2. Energy Measurement (CodeCarbon)

Energy Consumed	0.000010 kWh
CO2 Emitted	0.0036 g
Execution Time	0.0064 s
Measurement Mode	TDP-estimation (Fallback)

3. Real RAPL Benchmark (Energy-Languages Dataset)

Language	Energy (J)	Time (s)	Memory (MB)
Python	8920.5000	38.3000	124.5000
C	112.4000	1.8000	4.2000
C++	118.7000	1.9000	4.5000
Java	780.3000	6.2000	312.0000

4. SCI Scores

Estimated SCI	0.0036 gCO2eq/run
Real SCI	1.2463 gCO2eq/run
Deviation	99.71%
Anomaly	True

5. Green Score

Overall	75 / 100
Performance	72
Energy	67
Carbon	78.152461475
Maintainability	84.0

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6. Carbon Cost Projection

Per Run	0.0036 g CO2
Daily Runs (assumed)	500
Yearly CO2 (current)	0.6500 kg
Yearly CO2 (optimized)	0.4200 kg
Potential Savings	35%

Planner Reflection Note

Telemetry flagged: SCI deviation is 99.7%, exceeding the $\pm 15\%$ hardware benchmark variance limit.